

50 SQL Server DBA Interview Questions (TCS)

1. What is SQL Server?

SQL Server is Microsoft's relational database system used to store, manage, and retrieve data. In organizations, it's mainly used to handle application databases where we store structured data in tables and access it using SQL queries.

2. What is the default port number of SQL Server?

The default port used by SQL Server is 1433.

This is mainly used when we install the default instance.

3. Can we change the SQL Server port?

Yes, we can change it.

We do it through SQL Server Configuration Manager. Under SQL Server Network Configuration, we open TCP/IP settings and modify the port number in the IP Addresses section. After that, we restart the SQL Server service for the change to take effect.

4. What are the different SQL Server editions?

The common editions are:

- Enterprise Edition
- Standard Edition
- Developer Edition
- Express Edition
- Evaluation Edition
- Enterprise is used in large enterprises, while Express is mostly used for small applications.

5. Difference between Enterprise and Standard edition?

Enterprise edition supports advanced features like Always On availability groups, online indexing, and larger hardware limits.

Standard edition has some limitations in terms of features and scalability, but it's cheaper and works well for small to medium workloads.

6. What are SQL Server services?

The main services we usually deal with are:

SQL Server Database Engine

SQL Server Agent

SQL Server Browser

SQL Server Integration Services

The Database Engine handles the database operations, and SQL Server Agent is used for scheduling jobs like backups.

7. What is SQL Server Agent?

SQL Server Agent is a service used to schedule and automate tasks.

For example, we use it to schedule:

database backups

index maintenance

database integrity checks

data transfer jobs

8. What are system databases in SQL Server?

There are five main system databases:

- master
- model
- msdb
- tempdb
- resource (hidden)

These databases are created automatically when SQL Server is installed.

9. What does the master database store?

The master database stores server-level information like:

logins

configuration settings

database details

If the master database becomes corrupted, SQL Server may not start properly.

10. What is tempdb used for?

TempDB stores temporary objects like:

temporary tables

sorting operations

intermediate query results

It's heavily used by SQL Server internally.

11. What is the model database?

The model database acts as a template.

Whenever a new database is created, SQL Server copies the structure of the model database.

12. What is msdb used for?

MSDB stores information related to:

SQL Server Agent jobs

backup and restore history

maintenance plans

alerts

13. What is the difference between Login and User?

A Login is created at the server level.

A User is created inside a database.

So first we create a login, then we map that login to a database user.

14. What authentication modes exist in SQL Server?

Two types:

Windows Authentication

SQL Server Authentication

Windows authentication uses Active Directory credentials.

15. Which authentication mode is more secure?

Windows authentication is considered more secure because it uses Active Directory security policies and avoids storing passwords inside SQL Server.

16. What is Auto Growth?

Auto Growth allows database files to grow automatically when they run out of space.

For example, if the database file reaches its maximum size, SQL Server automatically increases the file size based on the configured growth setting.

17. What are recovery models?

SQL Server has three recovery models:

Simple

Full

Bulk Logged

They control how transaction logs are managed and how we recover databases.

18. What is Simple recovery model?

In Simple recovery, transaction logs are automatically cleared, so we cannot restore the database to a specific point in time.

It's usually used for non-critical databases.

19. What is Full recovery model?

In Full recovery model, every transaction is logged.

This allows point-in-time recovery, which means we can restore the database to an exact moment before a failure.

20. What is Bulk Logged recovery model?

Bulk Logged is similar to Full recovery but reduces logging for bulk operations like bulk insert or index rebuild.

Backup and Restore

21. What types of backups are available?

Common backups are:

Full Backup

Differential Backup

Transaction Log Backup

22. What is a Full Backup?

A full backup takes a complete copy of the entire database including data and part of the transaction log.

23. What is a Differential Backup?

A differential backup stores only the data that has changed since the last full backup.

24. What is Transaction Log Backup?

Transaction log backup records all transactions since the previous log backup.

This helps in point-in-time recovery.

25. How do you verify a backup?

We can use commands like:

RESTORE VERIFYONLY

This checks whether the backup file is valid without restoring it.

Performance and Indexes

26. What is an index?

An index helps SQL Server retrieve data faster by creating a structured path to the data instead of scanning the whole table.

27. What is a clustered index?

A clustered index determines the physical order of data in a table.

A table can have only one clustered index.

28. What is a non-clustered index?

A non-clustered index stores pointers to the actual table rows.

A table can have multiple non-clustered indexes.

29. What is an execution plan?

An execution plan shows how SQL Server executes a query internally.

It helps us understand where the query is slow.

30. How do you identify slow queries?

We can check:

execution plans

SQL Profiler

Query Store

DMVs

These tools help identify queries consuming high resources.

High Availability

31. What is Log Shipping?

Log shipping automatically sends transaction log backups from the primary server to a secondary server and restores them there.

This keeps the secondary database almost up to date.

32. What is Database Mirroring?

Database mirroring keeps a mirror copy of a database on another server.

If the primary server fails, the mirror server can take over.

33. What is Replication?

Replication copies data from one database to another database.

It is commonly used for reporting or distributing data across multiple locations.

34. What is Always On Availability Group?

It is a high availability solution that allows multiple replicas of a database across servers.

If one server fails, another replica can automatically take over.

35. What is a join?

A join is used to combine data from two or more tables based on a related column.

36. Types of joins?

Common joins include:

Inner join

Left join

Right join

Full outer join

37. What is a stored procedure?

A stored procedure is a group of SQL statements saved in the database that can be executed whenever needed.

38. What is a trigger?

A trigger is a special stored procedure that runs automatically when an event occurs like INSERT, UPDATE, or DELETE.

Practical DBA Questions

39. What will you do if the database is in Suspect mode?

First I check the SQL Server error log to understand the reason.

Then I try steps like:

setting database to emergency mode

running DBCC CHECKDB

repairing the database if possible

40. What will you do if the transaction log is full?

Steps I would take:

check if log backups are running

take a transaction log backup

check disk space

shrink the log file if necessary

41. What is DBCC CHECKDB?

DBCC CHECKDB checks the logical and physical integrity of the database and helps detect corruption.

42. What is RPO?

RPO stands for Recovery Point Objective.

It defines how much data loss is acceptable.

43. What is RTO?

RTO stands for Recovery Time Objective.

It defines how quickly the system must be restored after a failure.

44. What is a database snapshot?

A database snapshot is a read-only copy of a database at a specific point in time.

45. What is a deadlock?

A deadlock occurs when two transactions are waiting for each other's resources and neither can proceed.

46. How do you monitor SQL Server performance?

We can monitor using:

Performance Monitor

SQL Profiler

DMVs

Activity Monitor

47. What are DMVs?

Dynamic Management Views provide internal server information like memory usage, query performance, and session details.

48. What is TempDB contention?

It happens when multiple processes try to access TempDB resources simultaneously, causing performance issues.

49. What is a maintenance plan?

A maintenance plan automates routine tasks like:

backups

index rebuilds

database integrity checks

50. What is a database refresh?

Database refresh means copying data from production to another environment like testing or staging.

It is usually done by taking a production backup and restoring it on another server.